Lecture plan · for the lecturer

Adversaries and noise

University, vision 4

Time
45 min

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- Compare an autoencoder, a GAN and diffusion in one table.
- Name each method's typical failure.
- Explain why a sample is not evidence.

Activity: The table first

1. Warm-up (5 min): Show one generated picture and ask who believes it shows something real.
2. Table (15 min): Fill three rows on the board: method, loss, typical failure.
3. Lab (10 min): Pairs set the noise in the middle, then all the way up, and read the difference before and after the one step.
4. Cases (10 min): Students match three picture stories to a method and its failure.
5. Exit (5 min): Students name the method that removes noise and say why its sample proves nothing.

Discussion

- Which failure is whose?

Exit ticket

- Name the method that undoes noise.

Try it in your next lecture
Bring a generated image from the news and ask which method's failure it might be. Do not grade the guess as fact.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **Which idea removes a little noise at a time?**

◯ Diffusion ◯ A single 3×3 edge filter ◯ A GAN's discriminator

2 **A picture from a generator is evidence of what it shows.**

◯ True ◯ False

3 **Match the generator.**

Autoencoder	• • • •	Undo noise one step at a time
GAN	• • • •	Rebuild from a short code
Diffusion	• • • •	One network tries to fool another

4 **Order a diffusion story.**

☐ Learn a step that removes a little

☐ Repeat the step

☐ Add noise until the picture is gone

Answers · for the lecturer

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1 **Diffusion**

Diffusion repeats a small denoise. One step in the lab is the idea, not a finished generator.

2 **False**

It is a sample from a model trained to lower a loss.

3 **Autoencoder — Rebuild from a short code**
GAN — One network tries to fool another
Diffusion — Undo noise one step at a time

Three different losses. The lab only shows one denoise step.

4 **Top to bottom: 2, 3, 1**

The lab stops after one step, on purpose.

For you to think about

What will you say when a student treats a sample as evidence?